

Electoral Equilibrium: A Two-Page Summary

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July 13, 2026

The problem

Political shocks — scandals, endorsements, economic surprises, institutional rulings — do not move the electorate uniformly. Different demographic blocs react differently, so after a shock a party’s *optimal* coalition emphasis shifts. Electoral Equilibrium asks a single question: given a hypothetical shock described in plain text, what race-bloc coalition weighting best keeps a party above its win threshold, with what probability does that coalition win, and is the estimate partisan-neutral? The hard constraint is data scarcity: with only about twenty U.S. presidential cycles (1948–2024), the shock-to-response mapping cannot be regressed directly from history, so uncertainty must be modeled explicitly rather than assumed away.

The approach and pipeline

The system is a three-stage pipeline over three parallel demographic strata — race (5 blocs), religion (7), and gender (3), each independently covering the whole electorate, deliberately without cross-tabulation.

Stage 1 — language model. A Mistral-7B-v0.3 model, fine-tuned with QLoRA (rank 16, $\alpha=32$, 4-bit NF4, learning rate 2×10^{-4} , 3 epochs), maps a shock description to per-bloc loyalty shifts across all 15 blocs. Output is emitted as one of nine ordinal “delta bins” (**strong_neg** through **strong_pos**) via constrained decoding (the **outlines** library with a Pydantic schema), which guarantees every token is schema-valid. Held-out mean absolute error in delta units was **0.0362** on the *pre-rebalance Mistral v2* model (111-example eval) [**TODO: re-run MAE on the current v3 model against the 246-example eval set — 0.0362 is NOT the current figure**].

Stage 2 — optimizer. A disciplined quasi-convex program maximizes win probability in Sharpe-ratio form, $\max_w \Phi((\mu_{\text{eff}}(w) - V_{eq})/(\lambda_1 \sqrt{w^\top \Sigma_\Delta w}))$, where μ_{eff} is a λ -weighted blend of the three strata and V_{eq} is an electoral-college-adjusted win threshold (0.5066 Democrat, 0.4934 Republican). The five race-bloc weights are the decision variables, bounded to $[0.5 \times, 1.5 \times]$ their baseline population shares for demographic plausibility; the objective is provably quasi-convex and solved as a second-order cone program. Minimum-variance optimization is deliberately *not* used — it zeroes out small blocs, which is mathematically valid but politically nonsensical.

Stage 3 — Monte Carlo. Small-sample uncertainty is propagated with a Logistic-Normal isometric-log-ratio (ILR) simulation: coalition weights are mapped to unconstrained space via a Helmert contrast, the race-level covariance Σ_Δ is pushed through the ILR Jacobian, 10,000 draws are taken, and each is mapped back through a softmax to a valid coalition. The result is a win-probability point estimate with a 5th/95th-percentile confidence interval. Σ_Δ is a race-only 5×5 matrix estimated

by Ledoit–Wolf shrinkage over **9 cycles (1992–2024, 8 first-differences)**; pre-1990 cycles are excluded because thin blocs (`asian`, `latino`, `other_race`) had sparse or imputed sampling that more than doubled their apparent variance.

Data. Two distinct demographic inputs feed the model and must not be conflated. Per-bloc *loyalty* — how Democratic each bloc votes — is a multi-source survey merge over NEP exit polls, ANES, GSS, and CES, each first aggregated with its own sampling weights and then reconciled across sources by inverse-standard-error weighting (Pew NPORS and the Democracy Fund VOTER panel are wired into the loader but deferred), with a bloc-specific 1992 Perot two-party correction. Electorate *composition* — how large each bloc is, used to form the baseline coalition weights — comes separately from NEP exit-poll shares. One cycle is a known weak spot: ANES 2020’s sampling-weight column is null, so it falls back to equal-weighting (inflating the White Democratic share by ~ 3.6 pp) and is down-weighted or excluded from the affected merges. Supervision signal also comes from social media, tiered by how reliably a user’s demographics can be inferred: individual-bio posts feed both the mean and the covariance, while group- and platform-level proxies feed the mean only, and pure language-fallback posts are held out entirely — a discipline that protects the optimizer’s mean/variance-independence assumption. Because historical shock–response labels are scarce, the fine-tuning corpus is augmented with synthetic events generated by DeepSeek and reviewed by Gemini over a balanced taxonomy of event type, valence, electoral effect, and party. An earlier, label-skewed version (Mistral v2 era) was rebalanced and the model retrained; the current merged corpus is **1,228** records (`data/finetune/synthetic.jsonl`), split 982 train / 246 eval (`train.jsonl`, `eval.jsonl`), verified by line count.

Headline contribution: neutrality validation

The project’s strongest result is a protocol for proving the system is partisan-neutral — and using it to find and fix a real bias. A **zero-shock null test** feeds all-zero deltas through the full pipeline for both parties; a neutral system must return symmetric baselines. It did not: the Democratic baseline showed $\mu_{\text{eff}} = 0.5575$ (margin $+0.0509$, win probability 1.00) against the Republican 0.5000 (margin $+0.0066$, win probability 0.26). The cause was not the machine-learning model but an **equal-within-stratum weighting placeholder** that over-weighted small, strongly-Democratic religion and gender blocs by $7\text{--}33\times$. Replacing it with real electorate marginals cut the structural asymmetry from $+0.052$ to $+0.001$ — a roughly **98% reduction** — with the tiny residual now slightly favoring Republicans, exactly as the EC-adjusted threshold intends.

A **matched-pair symmetry test** (the same shock framed for each party) then exposed a smaller residual behavioral lean of $+0.028$ toward Democrats. This was traced by elimination: baseline aggregation was fixed; the Monte Carlo was switched to the real covariance; and the synthetic corpus was rebalanced to a -0.0007 label asymmetry and the adapter retrained — yet the lean persisted at $+0.028$. The conclusion is that the residual comes from the base model’s pretrained political priors, which survive light LoRA fine-tuning and concentrate in valence-interpretation events (e.g. a Republican gaffe read as more damaging than its Democratic mirror). It is disclosed as a limitation rather than tuned away, and a controlled comparison against a **Gemma-2-9b** model (identical corpus and hyperparameters) is set up to test whether the lean is base-model specific [**TODO: Gemma comparison run pending**].

Honest limitations

The three strata are combined additively without cross-tabulation, so the model inherits the ecological fallacy (Robinson 1950); μ_{eff} and the popular-vote threshold coincide only under that approximation. Only the five race blocs are rebalanced, with religion and gender held fixed and treated as neutral in the simulation, which understates uncertainty. The win threshold is a model-internal effective-loyalty quantity, not a literal vote count. Synthetic labels are LLM-generated, a documented deviation from the project’s reproducibility rule for this stage. And even after shrinkage the covariance rests on only eight first-differences, so the thinnest blocs remain data-limited. The residual +0.028 base-model lean is reported, not hidden.

Status

The pipeline runs end to end and is deployed behind a streaming web interface; the neutrality-validation finding is the paper’s central methodological contribution. Remaining work is final-number consolidation (fine-tuning MAE, the full symmetry table, and the Gemma comparison), IPF raking to replace the fixed religion/gender weights, and a covariance sensitivity analysis. **[TODO: insert final results table once runs complete.]**